

Hang Four Shirts on a Garment Rack SOP

(1x Episode: 4 Shirts)

One episode hangs four shirts on the garment rack. Each shirt is lifted off the shirt pile by its **left shoulder point** with the left gripper and held hanging in the air at the center of the table. The right gripper takes one hanger and pushes it down into the shirt through the collar. The left gripper pushes the left shoulder onto the left side of the hanger, and the right gripper puts the right shoulder onto the right side. The right gripper then carries the shirt to the rail by its hold on the right shoulder and sets the hook on the rail. The top button is closed with the shirt hanging on the rail, before the next shirt comes off the pile.

The order never changes: lift and hold, take the hanger, fit the hanger, hang, button the top button. One shirt is off the pile at a time. No shirt is left on the rail with a shoulder off its hanger end or with the top button open. If a shoulder slips off its hanger end or the hanger sits crooked, either gripper may set it right at any time. That is an adjustment, not a step.

Nothing is laid on the tabletop. The shirt goes from the pile into the air, from the air onto the hanger, and from the hanger onto the rail. Every hanger goes in through the **collar mouth** of a shirt held still by the left gripper. Nothing is pushed against the collar and nothing is forced.

The rack never moves. The top button is worked on the shirt as it hangs on the rail, and a gripper lets go only once the shirt is safely on the hanger or on the rail.

The left gripper takes every shirt off the pile and holds it at the fit spot while the hanger goes in. The right gripper does the primary work: it takes every hanger, pushes it down into the collar, puts the right shoulder on the hanger, carries the shirt to the rail and hangs it, and works the buttonhole over the button. The left gripper supports: it holds the shirt for the fit, pushes the left shoulder onto the hanger, and holds the top button for the buttoning. Both grippers work together on the button as needed.

The table is set up in one of three ways. Only the shirt pile moves; the rack, the fit spot, and the hanger pile are in the same place in all three. The shirt pile always sits where the left arm reaches it.

- **Config L1:** the shirt pile is in the back-left corner, left of the rack's left upright.
- **Config L2:** the shirt pile is in the front-left corner.
- **Config M:** the shirt pile is at the front-center, on the center line at the front edge.

Where a step depends on the setup it says so on an **IF** line: look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the shirt pile starts at the back-left (**Config L1**), the front-left (**Config L2**) or the front-center (**Config M**). One config per episode, chosen before recording and never changed mid-episode.
- **Left-arm rule:** the left gripper takes every shirt off the pile in every config. The pile is never staged on the right, and no arm reaches across the table for a shirt.
- **Fixed roles:** everything else is the same in all three configs. The right gripper takes every hanger from the hanger pile at the front right, puts it in, puts the right shoulder on, hangs the shirt, and works the buttonhole. The left gripper holds the shirt for the fit, puts the left shoulder on, and holds the top button.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the garment rack across the back, the shirt pile in the start zone for this episode's config, the hanger pile at the front right, and the clear center of the table where each shirt is held.
3. The rail is in frame end to end, from its **left end** to its **right end**, with clear space above it and below it, so every hook can be seen landing and the collar, both shoulders, and the top button of every hanging shirt can be seen.
4. The **fit spot** is in frame from the collar of a held shirt down to its hem.
5. Both arms are at home with grippers open.
6. The tabletop holds nothing except the garment rack, the shirt pile, the hanger pile, and robot hardware.
7. Nothing blocks the paths between the shirt pile, the hanger pile, the fit spot, and the rail.
8. The right arm reaches the whole hanger pile, the fit spot, and the whole rail from its left end to its right end. It does not lean out or stretch to a joint limit to get there.
9. The left arm reaches the whole shirt pile in every config, the fit spot, and the whole rail from its left end to its right end. It does not lean out or stretch to a joint limit to get there.

Materials checklist

1. One **garment rack** stands across the back of the table, in the middle of the table's back edge. Its **left upright** stands at the back left of center and its **right upright** at the back right

of center, so the **rail** runs left to right across the back.

2. The rack stands firm on its own base. It does not rock, slide, or lift when a hanger is set on the rail or a shirt is worked on it, and no gripper ever holds it up.
3. The rail is bare, straight, and clear along its whole length. It is high enough that a shirt on a hanger hangs with its hem clear of the tabletop and the shirt pile, and low enough that both grippers reach the collar, both shoulders, and the top button of a hanging shirt.
4. The rail is long enough that all four hangers hang on it in **one run**. Shirts may touch each other on the rail.
5. Four **shirts** sit in one **shirt pile** in the start zone for this episode's config. Each one lies **folded in half** and **face up**, collar toward the far side, with both **sleeves stretched flat**, laid straight on the one below it.
 - **Config L1:** back-left corner, left of the rack's left upright

Config L2: front-left corner

Config M: front-center, on the center line at the front edge

1. Every shirt has every button undone and its front open. No shirt is buttoned or rolled.
2. Every shirt is staged with its buttons on the left as seen from above and its buttonholes on the right, so the **left shoulder point** is on the left of the pile where the left gripper comes down on it.
3. Every shirt's **collar** opens wide enough, once the shirt hangs, that a hanger passes down through it with clear space around the hanger. A shirt whose collar will not open that far is not staged.
4. Every shirt's **top button** is sound and its **top buttonhole** is loose enough that the button passes through on one draw. A shirt with a loose, cracked, or missing top button is replaced before the episode.
5. Every shirt is dry, clean, and pressed, with no tears and no open seams.
6. Four **hangers** lie flat in one **hanger pile** at the front right, each laid straight on the one below it. Every **hook** points toward the far side with its **hook opening** facing the back.
7. All four hangers are the same shape and size, with both **shoulder ends** sound and a hook that is not bent. A hanger that will not sit straight on the rail is replaced before the episode.
8. Before collection, check the whole load fits: all four shirts hanging in one run on the rail, none on an upright and none with its hem on the tabletop.

Workspace layout

- **Garment rack:** across the back of the table, in the middle of the back edge. It holds the **rail**, the **left upright** at the back left of center, and the **right upright** at the back right of center. The rack stays where it is for the whole episode and no gripper ever lifts it.

- **Rail:** the bar the hangers go on. Its **left end** is the end toward the left of the table and its **right end** is the end toward the right. The first shirt of the episode goes at the left end and each shirt after it goes to the right of the last one.
- **Fit spot:** the air above the center of the table, just in front of the rail, where the left gripper holds each shirt while the hanger goes in. Both arms reach it. The tabletop under it stays bare.
- **Shirt pile:** the four shirts, in the start zone for this episode's config: the back-left corner (**Config L1**), the front-left corner (**Config L2**), or the front-center (**Config M**). The left arm can reach it.
- **Hanger pile:** at the front right, where the right arm can reach. The four hangers lie flat in one stack. The top hanger is always the next one.

The hangers are always staged on the right and the right arm takes every one. Every shirt is taken by the left arm and goes straight up to the fit spot at the center, so nothing is carried across the table.

The rack, the fit spot, and the hanger pile stay the same for every episode; only the shirt pile moves with the config.

Arm assignments

- **Right gripper:** works. It takes each hanger by the hook from the top of the hanger pile, pushes it down through the collar mouth, lets go of the hook, puts the shirt's right shoulder onto the right side of the hanger, carries the shirt by that hold to the rail and hangs it, and works the hole placket over the top button. It never takes a shirt off the pile.
- **Left gripper:** supports. It takes each shirt off the pile by the left shoulder point, holds it still at the fit spot while the hanger goes in, pushes the left shoulder onto the left side of the hanger, lets go once both shoulders are on the hanger, and holds the top button for the buttoning. It never takes a hanger.
- Wherever the left gripper holds, the right gripper works clear of it. The two grippers never crowd the same spot.

Vocabulary

- **Shirt:** one button-front shirt, hung whole. There are four in an episode and they are worked one at a time.
- **Shirt pile:** the four shirts stacked in the start zone: the back-left corner (**Config L1**), the front-left corner (**Config L2**), or the front-center (**Config M**). One per episode, chosen before recording and never changed mid-episode.
- **Face up:** the shirt lies with its front showing and its back against the pile. Every shirt is staged face up, so the left shoulder point is on top and ready to pinch.

- **Folded in half:** the shirt lies face up on the pile folded once across its middle, hem folded up toward the collar, collar toward the far side, with both sleeves stretched out flat and straight to the sides. The fold opens by itself when the shirt is lifted and hangs.
- **Collar:** the band and points around the neck opening.
- **Left shoulder point:** the top corner of the shirt where the left shoulder seam meets the sleeve, on the left as seen from above with the shirt face up. It is the part of the shirt the left gripper pinches to lift and hold it. The **right shoulder point** is the matching corner on the right.
- **Held shirt:** a shirt hanging from the left gripper's pinch on the left shoulder point, front facing the front edge, the fold fallen open, both fronts hanging open, both sleeves hanging down, and the hem hanging free and clear of the tabletop and the shirt pile.
- **Fit spot:** the air above the center of the table, just in front of the rail, where the left gripper holds each shirt for the hanger. Both arms reach it.
- **Collar mouth:** the neck opening of a held shirt, standing open just to the right of the left gripper's pinch, between the back of the collar and the two open fronts. A hanger goes down through this and nowhere else.
- **Forced fit:** the hanger pressed against the collar, so the collar pulls out of shape, a collar seam goes taut, or the collar band drags along with the hanger. Every fit goes straight down through an open collar mouth with clear space around the hanger. Lifting the hanger back out and bringing it down again is not a forced fit.
- **Retry:** a gripper opens, backs off, and takes the same grasp or makes the same fit again. A grasp that slips, a hook that slides out of the gripper, or a hanger that does not drop straight in is retried as many times as needed.
- **Hanger:** the frame the shirt hangs on. It has two **shoulder ends** and one **hook**.
- **Hanger pile:** the four hangers stacked flat at the front right, hooks toward the far side and hook openings facing the back. The top hanger is always the next one.
- **Hook:** the curved top of the hanger. It is the only part of the hanger a gripper ever touches, and it is the only part that ever sits on the rail. Once the shirt is on the hanger, the grippers hold the shirt, not the hook. Its **hook opening** is the gap under the curve, and it faces the back on every hanger.
- **Shoulder end:** the tip at either end of the hanger. The **left side of the hanger** ends in the left shoulder end and the **right side** in the right shoulder end, as seen from the front edge.
- **Shoulder seam:** the seam across the top of the shirt where the sleeve joins the body. There is one on the left and one on the right.
- **Hole placket:** the strip down the right side of the shirt front, as seen from the front edge, that carries the buttonholes. The buttons are on the left side of the front.
- **Top button:** the button closest to the collar. It is the only button worked in the episode. **Top buttonhole:** the buttonhole closest to the collar on the hole placket.

- **Hold the button:** the left gripper pinches the top button and keeps hold of it while the buttonhole is worked over it, until the button is through.
- **Buttoned:** the whole button sits flat on top of the hole placket with its whole rim clear of the buttonhole, and it stays there when the gripper lifts away. The two collar points come together and the hook stands up between them.
- **Lift away:** raise the item straight up until there is daylight under it, before anything moves sideways.
- **Carry level:** move the item across at one height, turned the same way as when it was picked up, without turning, swinging, or shaking it.
- **One run:** all four hangers sit on the rail in a single line, in the order they were hung, left end first, with no shirt behind another, no hook on another hanger, and no shirt on an upright. Shirts in the run may touch or brush each other.
- **Settled:** the item stays put when the gripper lifts away, with nothing rocking, sliding, swinging off, falling, or springing open.
- **Let-go point:** at the fit spot once both shoulders are on the hanger, for the left gripper; over the rail with the hook already resting on the rail, for the right gripper; or on a hanging shirt with the cloth resting where it was laid, for either gripper. A gripper opens nowhere else.

Steps

Run Steps 1 to 5 four times, once per shirt. Then run Step 6 once. One shirt is off the pile at a time, and its top button is closed on the rail before the next one comes off the pile.

Only the carry in Step 1 depends on the config: the **left gripper** takes every shirt, and the IF line says which way it carries the shirt to the fit spot. Every other line is the same in all three configs.

Any grasp or fit may be retried. If a gripper slips off, a hook slides out of the gripper, or a hanger does not drop straight in, the gripper opens, backs off, and tries again, as many times as it takes. Nothing is ever forced to save a retry.

Setting a shoulder back on its hanger end, or straightening a hanger that sits crooked in the collar, is an adjustment, not a step. Do it only if needed, at any time before the next shirt comes off the pile, with whichever gripper is free.

Step 1: Lift the next shirt off the pile and hold it at the fit spot

Goal: the next shirt hangs from the left gripper at the fit spot, front facing the front edge, with its collar mouth open.

Look where the shirt pile is before reaching for the first shirt.

- **IF the shirt pile is at the back-left (Config L1):** the **left gripper** pinches the **left shoulder point** of the **top shirt of the shirt pile**, lifts it straight up until the whole shirt hangs clear of

the pile and the fold has fallen open, and carries it level to the right to the **fit spot**.

- **IF the shirt pile is at the front-left (Config L2):** the **left gripper** pinches the **left shoulder point** of the **top shirt of the shirt pile**, lifts it straight up until the whole shirt hangs clear of the pile and the fold has fallen open, and carries it level back and to the right to the **fit spot**.
- **IF the shirt pile is at the front-center (Config M):** the **left gripper** pinches the **left shoulder point** of the **top shirt of the shirt pile**, lifts it straight up until the whole shirt hangs clear of the pile and the fold has fallen open, and carries it level toward the far side to the **fit spot**.

Then, in all three:

- The **left gripper** holds the shirt still at the fit spot, front facing the front edge, hem hanging free and clear of the tabletop and the shirt pile, until Step 4 says it can let go.

Take only the top shirt, by the left shoulder point. Never pull one out from under another, never lift a shirt by a sleeve, the collar, a placket edge, or the hem. If the pinch slips or misses the shoulder point, the left gripper opens and takes it again.

Check: the shirt is a **held shirt**: it hangs by the left shoulder point, front facing the front edge, the fold fallen open, both fronts hanging open, both sleeves hanging down, the **collar mouth** standing open, nothing caught up over the collar, and the hem clear of everything. The pile holds the remaining shirts, still neat.

If the shirt has not fallen open, if a sleeve or a front is caught up over the collar, or if the shirt hangs twisted, the **left gripper** puts it back down on the pile, opens, and takes it again.

Step 2: Take the next hanger

Goal: one hanger in the right gripper, held straight above the collar of the held shirt.

- The **right gripper** closes on the **hook** of the **top hanger of the hanger pile** and lifts it straight up clear of the pile.
- The **right gripper** carries it level to the fit spot and holds it above the **collar mouth** of the held shirt, hook up.
- The **left gripper** holds the shirt still the whole time.

Take the top hanger only, by the hook. Never take a hanger by a shoulder end, never slide one out from under another, and never lift two at once. If the hook slides out of the gripper or the grasp slips, the **right gripper** opens and closes on the hook again, as many times as needed, until the hanger is held.

Check: the **right gripper** holds one hanger by the hook, hook up, above the collar mouth. The hanger pile holds the remaining hangers, still neat. The held shirt has not moved.

Step 3: Fit the hanger into the shirt

Goal: both shoulders on the hanger, with the left gripper holding the left shoulder and the right gripper holding the right shoulder.

- The **right gripper** brings the hanger and pushes it down through the **collar mouth**, placing it near the left gripper's pinch while keeping clear space around the hanger.
- The **right gripper** keeps putting the hanger in until the left side of the hanger is in a good position to enter the shirt's left shoulder.
- The **left gripper** pushes that side of the shirt toward the hanger, so that the left side of the hanger enters the left shoulder. The **left gripper** then keeps its hold on the shoulder, with the hanger inside it.
- The **right gripper** lets go of the hook, takes the shirt's **right shoulder point** (which is not yet on the hanger), brings it over to the right side of the hanger, and puts it on.
- Once both sides are on the hanger, the **right gripper** keeps its hold on the shirt's right shoulder.

The hanger goes in through the collar mouth and nowhere else. Never take it in through a sleeve or up from the hem. If it does not go in, lift it out and bring it down again.

Check: the shirt hangs on the hanger by both shoulders, with one shoulder end in each shoulder and the hook standing up out of the collar. The collar is not stretched or dragged out of shape. The **left gripper** holds the left shoulder and the **right gripper** holds the right shoulder.

If a shoulder end is in a sleeve or outside the shirt, the gripper on that side lifts that shoulder off and puts it on again. If the hook went down inside the shirt, the **right gripper** takes the hook, lifts the hanger back out through the collar mouth, and brings it down again. If the hook slides out of the right gripper, the **right gripper** closes on it again and carries on. There is no limit on these retries.

Step 4: Hang the shirt on the rail

Goal: the shirt hanging on the rail in one run, to the right of the shirt already there.

- The **left gripper** opens and moves away to the left. The **right gripper** carries the shirt level to the rail by the hold it has on the right shoulder, with the hanger inside, keeping the shirt clear of the shirt pile and any shirt already hanging.
- The **right gripper** sets the **hook** over the rail, **hook opening** facing the back, until the hook rests on the rail and the shirt hangs free. Then it opens and lifts away.
- Put the first shirt at the **left end** of the rail. Put each shirt after it to the right of the last one. The shirts may touch each other; no shirt goes behind, on top of, or over the hook of another.
- The **left gripper** stays clear of the rail until the right gripper is off the shirt.

Carry the shirt by the hold from Step 3. Never carry a shirt over one already hanging.

Check: the hook is fully over the rail with its opening facing the back, the shirt hangs free and **settled**, and it does not hang on an upright, the shirt pile, or the tabletop. All the shirts hung so far sit in **one run** from the left end, hooks all facing the same way. The rack has not moved.

If the hook is not fully on the rail, or if the shirt lands out of the run, the **right gripper** takes the hook, lifts the shirt off the rail, and hangs it again. Do not push or slide a hanger along the rail after letting go. If a shoulder has come off its hanger end, or the hanger sits crooked in the collar, either gripper lifts that shoulder and sets it on the end. This is an adjustment, not a step.

Step 5: Button the top button on the rail

Goal: the top button closed with the collar closed around the hook, the shirt still hanging on the rail.

Run 5.1, then 5.2, then 5.3.

5.1 Hold the button

- The **left gripper** pinches the **top button** and **holds the button**.

5.2 Lay the buttonhole over the button

- The **right gripper** pinches the outer edge of the **hole placket** in line with the **top buttonhole**, lifts it a little off the cloth behind it, carries it across to the left, and lays the **top buttonhole** directly on top of the **top button**.

5.3 Draw the button through

- The **right gripper** pinches the hole placket just past the far lip of the **top buttonhole** and draws it straight across to the left, until the whole button comes out through the hole and lies flat on top of the hole placket.
- Both grippers work together as needed: the **left gripper** holds the button while the **right gripper** draws, and either gripper may re-pinch, push, or pull the cloth until the button is through.
- Once the button is through, the grippers still on the shirt open and lift away.
- Go back to Step 1 for the next shirt. After the fourth shirt, go on to Step 6.

Work the top button only. The rest of the front stays open. Neither gripper pulls the shirt downward off the hanger.

Check: the top button is **buttoned**: the whole button sits flat on top of the hole placket with its whole rim clear of the buttonhole, and it stays there when both grippers lift away. The two collar points come together and the hook stands up between them. Both shoulders are still on their hanger ends and the hook still rests on the rail.

If the button slips back out, or if only part of its rim came through the hole, the **right gripper** lifts the hole placket back off the button, lets it hang open, and 5.2 runs again. If a shoulder came off its end during the button, either gripper sets it back on the end.

Step 6: End the episode

- Confirm all four shirts hang on the rail in one run from the left end, each with both shoulders on their hanger ends, the top button closed, hooks all facing the back, and nothing on an upright or the tabletop. Confirm the shirt pile and the hanger pile are both bare.
- Return both arms home with grippers open.
- Stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take each shirt off the rail by its hook, one at a time, lay it face up on the tabletop, unbutton the top button, and lift the hanger back out through the collar.
2. Stack the four hangers back into the hanger pile at the front right, flat, hooks toward the far side with the hook opening facing the back, each laid straight on the one below.
3. Smooth each shirt out and press it if it is creased or carries a shoulder mark from the hanger. A shirt that still shows a hanger mark is not staged again.
4. Rebuild the shirt pile in the start zone for the next episode's config: back-left corner (Config L1), front-left corner (Config L2), or front-center (Config M). Each shirt lies face up and folded in half, hem folded up toward the collar, collar toward the far side, both sleeves stretched flat, every button undone and the front open, buttons on the left and buttonholes on the right, each laid straight on the one below it.
5. Check every collar still opens wide enough, once the shirt hangs, for a hanger to pass with clear space around it, and that no collar is stretched, wavy, or pulled out of shape.
6. Check every top button is still firm and every top buttonhole still passes its button on one draw.
7. Check all four hangers: no bent hook, no cracked or split shoulder end, and each one still sits straight on the rail.
8. Check the rack stands firm, the rail is straight and clear along its whole length, and the uprights have not shifted.
9. Replace anything damaged: a shirt that is torn, stained, damp, or has a loose or missing top button; a shirt whose collar will not open or will not hold its shape; a bent hanger; or a rack that rocks or will not stand straight.
10. Wipe the tabletop and check the rack sits firm and has not shifted, and that the center of the table is bare.
11. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain several violations; tag each one separately. Keep the episode with its violation tags. Do not delete it just because a rule was broken.

Tag a violation only when something is forced, the order is broken, or the episode moves on with a fault left in place.

Violations

Violation: Config misaligned

- **Visible cue:** what the arms do does not match the config on the table: the shirt pile is not in the start zone for the config; the right gripper takes a shirt off the pile; a gripper reaches across the table for a shirt; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the left-arm rule (the left gripper takes every shirt off the pile in every config, no arm reaches across the table for a shirt, and the IF line followed is the one for the config on the table).
- **Coaching note:** look where the shirt pile is before the first reach, then follow that config's IF line through Step 1.

Violation: Wrong order, or two shirts worked at once

- **Visible cue:** the hanger comes over before the shirt is held at the fit spot; the shirt goes to the rail before both shoulders are on the hanger; the button is worked before the shirt is on the rail; or a second shirt or hanger comes off its pile before the shirt on the rail is buttoned.
- **SOP rule broken:** Steps 1 to 5, lift and hold, take the hanger, fit the hanger, hang, then button the top button, in that order, one shirt at a time.
- **Coaching note:** finish the shirt on the rail, button and all, before you touch the next one.

Violation: Wrong pickup from the shirt pile

- **Visible cue:** the left gripper takes a shirt other than the top one, works one out from under or between the others, slides one off sideways, lifts the pile to get at a lower shirt, lifts two shirts in one grasp, or shakes or swings a shirt to open it.

- **SOP rule broken:** Step 1, only the top shirt is taken, at the left shoulder point, and it is lifted straight up clear of the pile.
- **Coaching note:** come down on the top one, pinch the left shoulder point, and lift straight up. Nothing gets dug out.

Violation: Item taken by the wrong part

- **Visible cue:** a shirt is lifted off the pile by a sleeve, the collar, a placket edge, or the hem rather than the left shoulder point, and the episode carries on with that hold; a hanger is taken by a shoulder end rather than the hook; or a hanger other than the top one is taken, or two are lifted at once.
- **SOP rule broken:** Steps 1 and 2, the shirt comes off the pile at the left shoulder point, and every hanger is taken by the hook and always the top one of the pile.
- **Coaching note:** left shoulder point off the pile, hook for every hanger. Land on the grasp point or open and take it again.

Violation: Shirt not held at the fit spot

- **Visible cue:** the left gripper lays the shirt on the tabletop, drapes it over the rack or the pile, holds it off to one side instead of at the center in front of the rail, holds it with its hem dragging on the tabletop or the pile, holds it with its back to the front edge, or moves it about while the hanger is coming in.
- **SOP rule broken:** Steps 1 to 3, the left gripper holds the shirt still at the fit spot, front facing the front edge and hem hanging free, from the lift until both shoulders are on the hanger.
- **Coaching note:** lift it to the middle, face it to the front, and hold it still. The shirt never touches the table.

Violation: Hanger brought to a shirt that is not hanging right

- **Visible cue:** the hanger comes over while the shirt has not fallen open, hangs twisted, or has a sleeve or a front caught up over the collar, and the shirt is not put back on the pile and taken again first.
- **SOP rule broken:** Step 1, a shirt that has not fallen open, hangs twisted, or has something caught up over the collar is put back down on the pile and taken again before the hanger comes over.
- **Coaching note:** look at the shirt before you reach for the hanger. If it is not hanging right, put it back and take it again.

Violation: Collar stretched or hanger forced through

- **Visible cue:** the hanger presses against the collar going in, the collar pulls out of shape, a collar seam goes taut, the collar band drags along with the hanger, the hanger is pushed or wiggled through against the cloth instead of being lifted out and brought down again, or the hanger goes in through a sleeve or up from the hem instead of through the collar mouth.

- **SOP rule broken:** Step 3, the hanger goes straight down through the open collar mouth with clear space around it and is never pressed against the collar. A hanger that does not drop in is lifted out and brought down again.
- **Coaching note:** if it does not drop straight in, take it out and bring it down again. Never push against the collar.

Violation: Shirt carried with a shoulder off the hanger

- **Visible cue:** the right gripper carries the shirt toward the rail while a shoulder end is in a sleeve, outside the shirt, or not in its shoulder; the left gripper lets go before the left side of the hanger is in the left shoulder; or the right shoulder is never put onto the right side of the hanger.
- **SOP rule broken:** Step 3, the left side of the hanger goes into the left shoulder and the right shoulder is put onto the right side, and both are on before the shirt leaves the fit spot.
- **Coaching note:** both shoulders on the hanger before you go to the rail.

Violation: Hanger let go before it rests on the rail

- **Visible cue:** the right gripper opens above the rail and drops the shirt and hanger onto it; it lets go before the hook rests on the rail, so the shirt drops the last part of the way or swings off; or a gripper opens anywhere other than at a let-go point.
- **SOP rule broken:** Step 4, the right gripper opens only once the hook rests on the rail, and a gripper opens nowhere but at a let-go point.
- **Coaching note:** all the way down, then let go. Watch it as the gripper comes off.

Violation: Hanger not fully on the rail

- **Visible cue:** the hook rests on top of the rail rather than over it, hangs on the hook of the hanger beside it, hangs on an upright, or sits half on with the shirt tipping, and the episode moves on to Step 5 without the shirt being lifted off and hung again; or the first shirt of the episode is hung somewhere other than the left end, or a shirt is hung to the left of one already hanging, and the episode moves on.
- **SOP rule broken:** Step 4, the hook is brought down onto the rail until it rests over the rail with the shirt hanging free, the first shirt goes at the left end and each one after it to the right of the last, and a bad hang is fixed by lifting the shirt off by its hook and hanging it again before Step 5.
- **Coaching note:** seat the hook over the bar and watch it hold before you let go. Left end first, then rightward. If it is not right, lift it off and hang it again.

Violation: Hooks not all facing the same way

- **Visible cue:** a hook is set on the rail with its opening facing the front while the others face the back, or a hanger is turned on the rail after it is hung.

- **SOP rule broken:** Step 4, every hanger is set down on the rail with its hook opening facing the back.
- **Coaching note:** openings to the back, every hanger, as it goes on the rail.

Violation: Button not held

- **Visible cue:** the left gripper is not on the top button while the right gripper lays or draws the buttonhole; or the left gripper holds a collar point, the hole placket, or the body of the shirt instead of the top button.
- **SOP rule broken:** Step 5.1, the left gripper pinches the top button and holds it while the buttonhole is laid over it and drawn through.
- **Coaching note:** hold the button itself.

Violation: Buttonhole not laid over the button, or the button forced

- **Visible cue:** the right gripper pushes the button at a closed hole placket, shoves the two sides of the front together, or draws the hole placket across before the buttonhole is over the button.
- **SOP rule broken:** Steps 5.2 and 5.3, the top buttonhole is laid on top of the top button first, then the hole placket is drawn across until the whole button comes through.
- **Coaching note:** hole on the button first, then draw it through.

Violation: Top button left open, or the closure not confirmed

- **Visible cue:** the next shirt comes off the pile with the top button of the last one open; the button slips back out of the hole and the episode moves on; only part of its rim is through the hole; or the two collar points do not come together around the hook.
- **SOP rule broken:** Step 5.3, the whole button sits flat on top of the hole placket with its rim clear of the hole and stays there when both grippers lift away, before the next shirt is taken.
- **Coaching note:** look at the button after both grippers come off. If it slipped, open the placket and do it again.

Violation: Wrong button worked

- **Visible cue:** a button below the top one is closed, more than one button is closed, or a lower buttonhole is laid over the top button so the front runs crooked.
- **SOP rule broken:** Step 5, the top button and the top buttonhole are the only closure worked, and the rest of the front stays open.
- **Coaching note:** top button only, the one closest to the collar.

Violation: Hung shirt pushed or slid along the rail

- **Visible cue:** a gripper pushes, slides, or re-spaces a shirt already hanging on the rail instead of lifting it off the rail by its hook and hanging it again; a gripper adjusts the

shoulders or works the button of a shirt other than the one just hung; or a gripper reaches through the run to reach the rail.

- **SOP rule broken:** Steps 4 and 5, a bad hang is fixed by lifting the shirt off the rail by its hook and hanging it again, and only the shirt just hung is worked on the rail.
- **Coaching note:** lift it off and hang it again. Nothing gets pushed along the rail.

Violation: Rack moved or lifted

- **Visible cue:** the rack rocks, slides, tips, or lifts off the tabletop; the rail bends or twists under a hook or while a shirt is worked on it; or either gripper holds, drags, leans on, or turns the rack.
- **SOP rule broken:** Steps 4 and 5, the rack stands on its own base and never moves, and no gripper touches it.
- **Coaching note:** the rack stands on its own. If it is moving, stop and reseal it.

Violation: Arms swapped roles or crowded the same spot

- **Visible cue:** the right gripper takes a shirt off the pile or holds it for the fit; the left gripper takes a hanger or carries a shirt to the rail; the right gripper pushes the left shoulder onto the hanger, or the left gripper puts the right shoulder on; or the two grippers work the same spot, touch each other, or one has to move before the other can reach.
- **SOP rule broken:** Steps 1 to 5, the left gripper lifts and holds each shirt, puts the left shoulder on the hanger, and holds the top button, and the right gripper takes the hanger, puts it in, puts the right shoulder on, hangs the shirt, and works the buttonhole.
- **Coaching note:** left lifts and holds, right hangs and buttons. Neither one does the other's work.

Violation: Item dropped or something knocked over

- **Visible cue:** a shirt or a hanger falls onto the tabletop or the floor; a shirt swings off the rail or is knocked off it; a gripper strikes the rack or an upright and it shifts; the shirt pile or the hanger pile is knocked over or dragged out of place; or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 5, each item stays on a clear travel path, a shirt already hung is left hanging on the rail, and a gripper opens only at a let-go point.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with fewer than four shirts on the rail, a shirt on an upright, a shoulder off its hanger end, a top button open, something still on the shirt pile or the hanger pile, either arm away from home, a gripper closed, or an arm doing something else after homing.
- **SOP rule broken:** Step 6, confirm the four hung shirts, their shoulders, their buttons, and the bare piles, return both arms home with grippers open, then stop recording.

- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Bad shirt:** a torn shirt or open seam, a top button that comes off or is missing, a buttonhole that will not pass its button after three correct draws, or a collar that will not open wide enough for a hanger to pass with clear space around it. Replace it before the next episode.
- **Bad hanger:** a bent or broken hook, a cracked or split shoulder end, or a hanger that will not sit straight on the rail. Replace it before the next episode.
- **Rack fault:** the rack will not stand straight, an upright works loose, or the rail bends or comes out of its uprights under a hung shirt. Repair or replace it before the next episode.
- **Rail too short:** all four hangers will not fit on the rail in one run.
- **Rail at the wrong height:** a hung shirt's hem rests on the tabletop, or a gripper cannot reach the shoulders or the top button of a hanging shirt.
- **Rack shifts out of place** during the episode.

Annotation subtasks (from SOP)

1. Lift one shirt off the pile by the left shoulder point and hold it at the fit spot
2. Take one hanger from the hanger pile
3. Push the hanger down through the collar
4. Push the left shoulder onto the hanger
5. Put the right shoulder onto the hanger
6. Hang one shirt on the rail to the right of the last one
7. Lay the top buttonhole over the top button
8. Draw the top button through the hole
9. Return both arms home and end the episode